

# Home Tasks - 04

Student Name: ADITYA  
SHARMA  
Branch: CSE - AIML  
Semester: 5  
Subject Name: ADBMS

UID: 24BAI70379  
Section/Group: 24AIT\_KRG-1\_A  
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Subject Code: 24CSP-310

## Q1.

A phone call is considered an international call when the person calling is in a different country than the person receiving the call.

What percentage of phone calls are international? Round the result to 1 decimal.

Assumption:

- The **caller\_id** in **phone\_info** table refers to both the caller and receiver.

**phone\_calls** Table:

| Column Name | Type      |
|-------------|-----------|
| caller_id   | integer   |
| receiver_id | integer   |
| call_time   | timestamp |

**phone\_calls** Example Input:

| caller_id | receiver_id | call_time           |
|-----------|-------------|---------------------|
| 1         | 2           | 2022-07-04 10:13:49 |
| 1         | 5           | 2022-08-21 23:54:56 |
| 5         | 1           | 2022-05-13 17:24:06 |
| 5         | 6           | 2022-03-18 12:11:49 |

**phone\_info Table:**

| Column Name  | Type    |
|--------------|---------|
| caller_id    | integer |
| country_id   | integer |
| network      | integer |
| phone_number | string  |

**phone\_info Example Input:**

| caller_id | country_id | network  | phone_number    |
|-----------|------------|----------|-----------------|
| 1         | US         | Verizon  | +1-212-897-1964 |
| 2         | US         | Verizon  | +1-703-346-9529 |
| 3         | US         | Verizon  | +1-650-828-4774 |
| 4         | US         | Verizon  | +1-415-224-6663 |
| 5         | IN         | Vodafone | +91 7503-907302 |
| 6         | IN         | Vodafone | +91 2287-664895 |

**Example Output:**

| international_calls_pct |
|-------------------------|
| 50.0                    |

**Solution:**

```

SELECT
ROUND(100.0*COUNT (
CASE WHEN C.COUNTRY_ID<>R.COUNTRY_ID THEN 1
END
)/ COUNT(*), 1)
AS INTERNATIONAL_CALLS_PCT
FROM PHONE_CALLS AS P
JOIN PHONE_INFO AS C
ON P.CALLER_ID=C.CALLER_ID
JOIN PHONE_INFO AS R
ON P.RECEIVER_ID=R.CALLER_ID

```